

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE 2500: Engineering Probability, Spring 2026

Homework #4: due Thursday, Feb. 26th, at 2PM.

Show all work for full credit!

Submit your work as a single PDF on Gradescope, labeling each problem number with a page.

1. (25 points.) Let X be the number of minutes a student spends in an exam before realizing they forgot to bring a calculator. The CDF for the random variable X is described by the formula

$$F_X(x) = \begin{cases} 0 & x < 0 \\ \frac{3}{1600} (400 - (x - 20)^2) & x \in [0, 20] \\ \frac{1}{240}x + \frac{2}{3} & x \in [20, 80] \\ 1 & x > 80 \end{cases}$$

- (a) (5 points.) Sketch the CDF (by hand). Make sure to include important points on the x and y axes.
- (b) (5 points.) Use the CDF to compute $P(X < 10)$.
- (c) (5 points.) Use the CDF to compute $P(X \in [15, 40])$.
- (d) (10 points.) Determine the formula for the PDF corresponding to this CDF, and sketch it (by hand). Make sure to include important points on the x and y axes.

2. (20 points.) The PDF for a continuous random variable X is described by the formula

$$f_X(x) = \begin{cases} c(x^3 + 1) & x \in [-1, 1] \\ 0 & \text{otherwise} \end{cases}$$

- (a) (5 points.) Determine the value of c that makes this a valid PDF.
- (b) (5 points.) Sketch the PDF (by hand) for the random variable X . Make sure to include important points on the x and y axes.
- (c) (5 points.) Use the PDF to compute $P(X < 0)$.
- (d) (5 points.) Determine the CDF corresponding to this PDF.

3. (15 points.) The time between illegal “double touch” violations in Olympic curling is modeled by an exponential random variable X (measured in hours) with parameter $\lambda = \frac{1}{3}$.

- (a) (5 points) Compute the probability that there are no violations in a 2-hour match.
- (b) (5 points) Compute the probability that there are exactly 3 violations in a 6-hour tournament. Hint: remember the connection between the exponential and Poisson random variables.
- (c) (5 points) What would λ need to be so that the events {less than 1 hour between violations} and {greater than 1 hour between violations} are equally probable? That is, half of the PDF’s mass is on one side of 1 hour and half is on the other (the median, not the mean).

4. (20 points.) Let X be the time it takes for HUNTR/X to fly to their next concert. We model X as a Gaussian random variable with mean 6 hours and standard deviation 2 hours. Use the numerical Q table from the book (p. 169) or another source to estimate the following probabilities to 4 decimal places. You can also use an online Q function calculator, but keep in mind you will only have access to the printed Q table on the exams.

(a) (6 points.) $P(X > 5)$

(b) (6 points.) $P(|X - 6| > 0.6)$

(c) (8 points.) $P(X \in [3, 7.8])$

5. (20 points.) Now let Y be the total time it takes for HUNTR/X to finish eating their pre-show ramyeon, modeled as a Gaussian random variable with mean 30 sec and standard deviation 5 sec. Use the Q table (or an online inverse Q function calculator) to estimate the following values.

(a) (6 points.) The value a such that $P(Y > a) = 0.2$

(b) (6 points.) The value b such that $P(|Y - 30| < b) = 0.3$

(c) (8 points.) Compute the probability that it takes them more than 38 seconds to finish their ramyeon, given that they have already been eating for at least 32 seconds.